

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000844	Version:	8.0, CURRENT
	Status:	Effective	Effective Date:	19-Jul-2011
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DG-57 / DG-517 Service Manual

TECHNICAL SERVICE SOFTWARE OVERVIEW

Procedure description
This procedure describes how to use the Technical Service Software, both in normal mode and “NoInit” mode. “NoInit” mode is useful when an initialization problem occurs because in this mode the instrument is not initialized, but you get only access to the most basic functions. Note: Document REGD-0000844_v8 replaces document REGD-0000844_v7.

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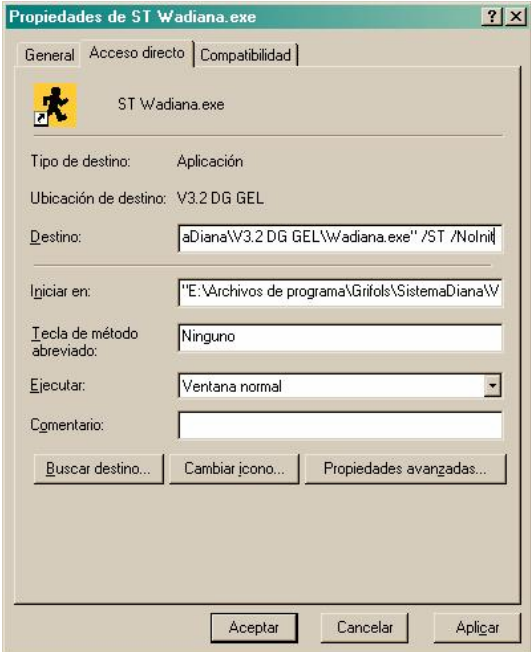
EQUIPMENT

1. An instrument connected with a PC with the software installed

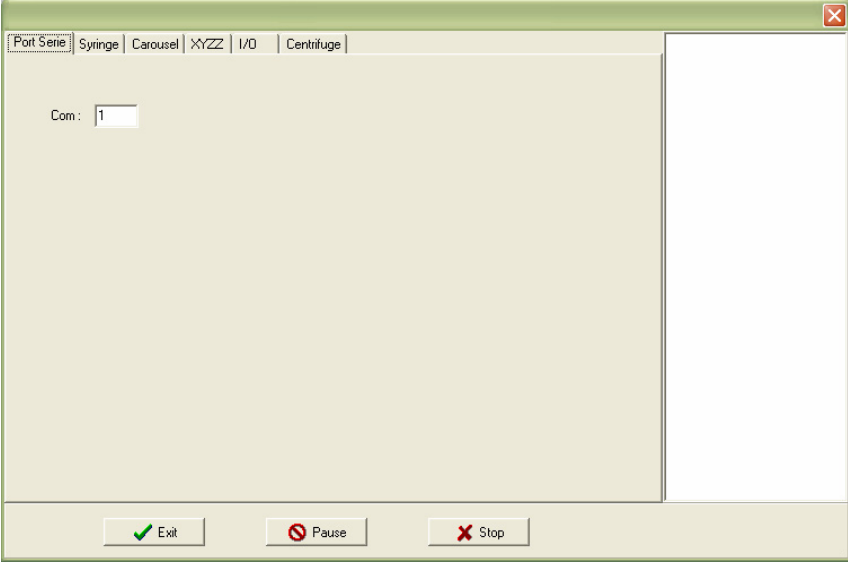
SET UP

1. Turn on the instrument.

PROCEDURE 1 – “Nolnit” mode

Process	Expected Results
1. Make a copy of the software shortcut on the desktop by a copy and paste. 2. Right click the new shortcut and select the “Properties” option. 3. Add the text “/ST /Nolnit” after the software path. Note this text is <u>case sensitive</u>. See Picture 1.	A new shortcut in the desktop is created. Picture 1 window appears. ---
 Picture 1	---
4. Press “Ok” button.	---

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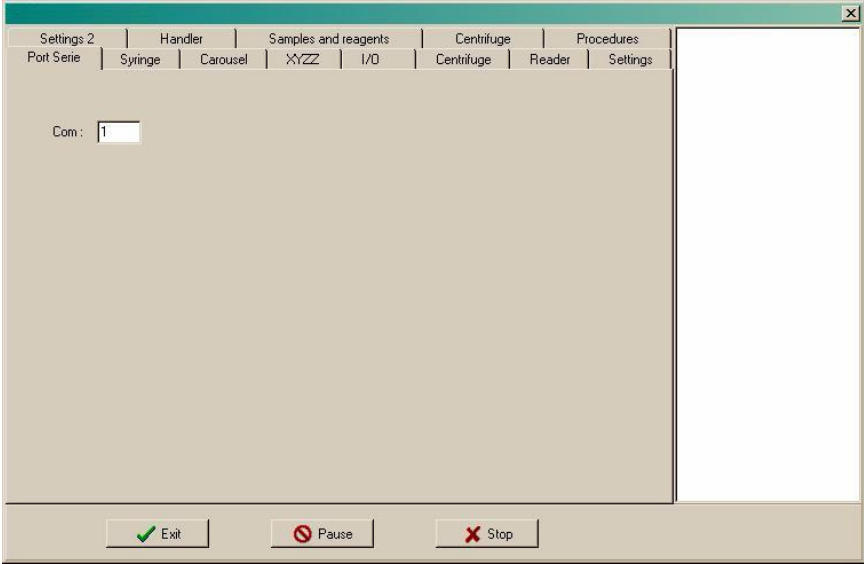
Process	Expected Results
5. Double-click the new created shortcut.  Picture 2	Technical service software starts under “NoIrit” mode and Picture 2 window appears.
6. Enter the Technical Service password. (“D+G_DIANA”) 7. Now you get access to “Port Serie”, “Syringe”, “Carousel”, “XYZZ”, “I/O” and “Centrifuge” tabs. See Annex A for and explanation of the functions on every panel. See picture 3.  Picture 3	--- ---
8. The right panel is intended for listing the error codes than occurs during ST software execution. See “Error codes list” document to know they meanings. 9. Because the software doesn’t initialize the instrument it may appear the following errors on the right panel when you select the corresponding tab: “CarrErrCod C”, “XYZErrCod 4”, “IOErrCod 4” and “CentErrCod C”. These errors mean that a board has been reset and will appear again every time you turn off the instrument. 10. To exit the software press the “Exit” button. (See Picture 3) 11. Don’t forget to delete the new shortcut from the desktop.	--- --- Program closes. ---

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PROCEDURE 2 – Normal mode

Process	Expected Results
1. A shortcut must be present in the “Start Menu” ended with the “ST” suffix. If not, you can create one shortcut following steps 2 to 5. Otherwise go directly to step 6. 2. Make a copy of the software shortcut on the desktop by a copy and paste. 3. Right click the new shortcut and select the “Properties” option. 4. Add the text “/ST” after the software path. Note this text is <u>case sensitive</u>. See Picture 4. Propiedades de ST Wadiana.exe General Acceso directo Compatibilidad  ST Wadiana.exe Tipo de destino: Aplicación Ubicación de destino: V3.2 DG GEL Destino: \\SistemaDiana\V3.2 DG GEL\Wadiana.exe" /ST Iniciar en: "E:\Archivos de programa\Grifols\SistemaDiana\W Tecla de método abreviado: Ninguno Ejecutar: Ventana normal Comentario: Buscar destino... Cambiar ícono... Propiedades avanzadas... Aceptar Cancelar Aplicar Picture 4 5. Press “OK” button. 6. Double-click the shortcut. 7. Enter the Technical Service password. (“D+G_DIANA”)	--- A new shortcut in the desktop is created. Picture 4 window appears. --- --- Technical service software starts under normal mode, instrument initialize and Picture 2 window appears. ---

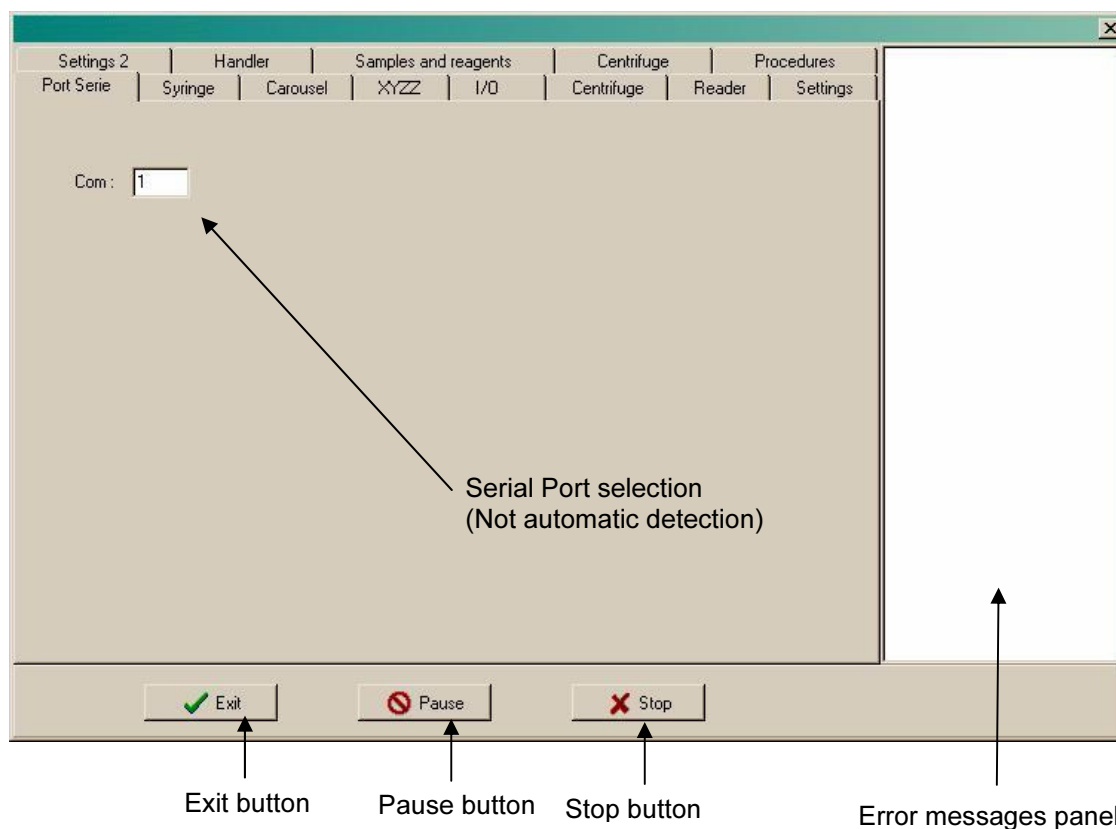
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Process	Expected Results
8. Now you get access to “Port Serie”, “Syringe”, “Carousel”, “XYZZ”, “I/O”, “Centrifuge”, ”Reader”, “Settings”, “Settings 2”, “Handler”, “Samples and reagents”, “Centrifuge” (High level) and “Procedures” tabs. See Annex A for and explanation of the functions on every panel. See picture 5.  The screenshot shows a software window with a title bar and a close button. Below the title bar is a row of tabs: 'Settings 2', 'Handler', 'Samples and reagents', 'Centrifuge', and 'Procedures'. Below these is another row of tabs: 'Port Serie', 'Syringe', 'Carousel', 'XYZZ', 'I/O', 'Centrifuge', 'Reader', and 'Settings'. The main area of the window is a large, empty, light-colored rectangle. In the bottom left corner of this area, there is a label 'Com:' followed by a small input field containing the number '1'. At the bottom of the window, there is a row of three buttons: a green checkmark button labeled 'Exit', a red circle with a slash button labeled 'Pause', and a red X button labeled 'Stop'. Picture 5	---
9. The right panel is intended for listing the error codes than occurs during ST software execution. Consult "Error codes list" document to know their meanings.	---
10. To exit the software, press the “Exit” button. (See Picture 5)	Program closes.
11. Don't forget to delete the new shortcut from the desktop if you had created it.	---

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Annex A – ST software Tabs explanation

“Port Serie” Tab.



As a general rule for all tabs:

- Use “Exit button” to leave the software.
- Use “Pause button” to stop temporarily an ongoing procedure. (This action may delay depending on the procedure)
- Use “Stop button” to end a procedure. (This action may delay depending on the procedure)
- The “Error messages panel” is used to print the error codes generated. (Consult “Error codes list”) document to know they meanings)

In the “Port Serie” Tab you can configure the serial port of where the instrument is connected. After change this value, exit the software and start again in order it takes effect.

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Syringe Tab

This Tab is used to send direct commands to the syringe module. It is useful to verify the syringe is responding to commands. For a full test of the syringe use the “Pipeting test” of the “Procedures” tab.

Buttons behavior:

Button	Behavior
Right setup	Initializes the syringe.
Absolute position	Syringe reaches the absolute position corresponding to selected (textbox) amount of steps (0..3000).
Aspirate	Aspirates the selected (textbox) amount of steps (0..3000).
Dispense	Dispense the selected (textbox) amount of steps (0..3000).
Input	Moves the valve to input position (to bottles).
Output	Moves the valve to output position (to probe).
Run	Runs the settled command.
Repeat last command	Repeats last command without necessity to settle and run it again
Stop	Stops immediately any running command.

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Carousel Tab

This Tab is used to send direct commands to the carousel module.

Buttons behavior:

Button	Behavior
Move sample carousel	Moves samples carousel to any count
Mover reagent carousel	Moves reagents carousel to any count
Samples home	Does a home of the samples carousel
Reagents home	Does a home of the reagents carousel
Idle	Put both carousel motors in idle state, so you can move them by hand.
Spinning reagents	Configures a spinning pattern for test the reagents spinning system.
Stop spinning	Stops spinning reagents
Activate peripheral	Activate led lights. Peripheral 3 is the card barcode light and peripheral 4 is samples and reagents lights.
Deactivate peripheral	Deactivate led lights. Peripheral 3 is the card barcode light and peripheral 4 is samples and reagents lights.
Update	Reads the samples and reagents encoders counts and displays them in "carousel" and "reagents" text boxes.
Reset	Operates a Carousel Board microcontroller's reset.
EEPROM	Displays Carousel Board EEPROM's values.

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XYZZ Tab

This Tab is used to send direct commands to the XYZZ module.

Buttons behavior:

Button	Behavior
Joystick	Moves any motor of the arm in small movements. Maintain pressed a button and the motor will move faster.
Move 3 motors	Moves 3 motors at a time to any count.
Move 2 motors	Moves 2 motors at a time to any count.
Move 1 motor	Moves 1 motor to any count.
Home	Does a home of a motor.
EEPROM	Displays XYZZ Board EEPROM's values.
Reset	Operates a XYZZ Board microcontroller's reset.
Card check	Check card presence.
Level reset	Resets level detector status.
Idle	Puts all four motors in idle state.
Gripper release	Opens gripper fingers.
Update	Reads the four motor encoders count and displays them.

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I/O Tab

This Tab is used to send direct commands to the I/O module.

Most of the indicators are for read only purposes:

Indicator	Meaning
Analog inputs	Readings from load cells and temperature sensors.
Alarm	Firmware alarms.
Digital inputs	Pressure sensors.
Status of incubators	Incubator on temperature.
Threshholds points	Information of temperature and load cells adjustments.

Button behavior:

Button	Behavior
Cont. Reading	Starts to continuous monitor and refresh all inputs.
Heating control	Activates/deactivates the heating control.
Temperature 1	Programs the temperature of heater block 1.
Temperature 2	Programs the temperature of heater block 2.
External wash	Activates the washing station valve (EVA).
Pumps	Activates pumps.
Internal wash	Activates the C valve (EVC).
Washing dilution cup.	Activates the B valve (EVB).
Open lid	Open instrument door.
Draining dilution cup	Activates the E valve (EVE).
EEPROM	Displays I/O Board EEPROM's values.
Reset	Operates an I/O Board microcontroller's reset.

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Centrifuge Tab (Low level)

This Tab is used to send direct commands to the centrifuge module.

Buttons behavior:

Button	Behavior
Position	Moves centrifuge to any count.
Home	Does a home of the centrifuge.
Idle	Put centrifuge motor in idle state, so you can move it by hand.
EEPROM	Displays Centrifuge Board EEPROM's values.
Reset	Operates a Centrifuge Board microcontroller's reset.
Random movements	Starts a cycle of random movements.
Update	Reads centrifuge encoder counts and displays them.

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Reader Tab

This Tab is used to work with the cameras module.

Buttons behavior:

Button	Behavior
Read Card 1	Reads a card in incubator first position with group algorithm.
Read Card 1-B	Reads a card in incubator first position with screening algorithm.
Show Picture	Displays the cameras pictures.
Read Sample	Reads a sample barcode and diameter.
Read reagent	Reads a reagent barcode.
Reagent BC check	Reads a reagent barcode continuously and displays failure statistics.
Reader sensibility	Change this to configure the instrument sensibility.

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Settings Tab

The screenshot shows the 'Settings Tab' window. It has a title bar with a close button. The window is divided into several sections, each with a tab-like header. The sections are: Settings 2, Handler, Samples and reagents, Centrifuge, Procedures, Port Serie, Syringe, Carousel, XYZZ, I/O, Centrifuge, Reader, and Settings. The 'Settings' section is active. It contains several sub-sections: Temperature of the incubators (with Start, Last setting, and n/a buttons), Tank level sensors (with Start, Last setting, and 18/6/2009 buttons), Centrifuge (with Start, Last setting, and 18/6/2009 buttons), Carousel (with Reagents carousel offset and Magnetic shaker checkbox), Handler (with Start, Last setting, and 30/6/2009 buttons), Gripper delay (with Start, Last setting, and 30/8/2007 buttons), Reader (with Reader reference and Save Values buttons), Camera alignment (with Reader, Samples, and Reagents buttons), Camera distance (with 148 input field), Camera optics (with Reader, Samples, and Reagents buttons), and Ser. N° (with 0128 input field). At the bottom, there are three buttons: Exit (green checkmark), Pause (red circle with slash), and Stop (red X).

This Tab is used to adjust the instrument settings.

Panels behavior:

Panel	Behavior
Temperature of the incubators	Adjusts incubators. Consult REGD-0000799.
Tank level sensors	Adjusts tank level sensors. Consult REGD-0000819
Centrifuge	Adjusts the gripper with respect the centrifuge. Consult REGD-0000808
Carousel	Adjusts carousel and magnetic shaker. Consult REGD-0000796 and REGD-0000797
Handler	Adjusts handler with respect the incubator. Consult REGD-0000807
Gripper delay	Adjusts gripper delays. Consult REGD-0000807
Reader	Does a reader image reference. Consult REGD-0000814
Save values	Saves settings in a file CalibValues.txt file (in Log folder). Overwrites existing CalibValues.txt file.
Camera alignment	Adjusts the cameras alignment. Consult REGD-0000810, REGD-0000811 and REGD-0000812
Camera distance	Adjusts the samples camera distance. Consult REGD-0000810
Camera optics	Adjusts the cameras optics. Consult REGD-0000810, REGD-0000811 and REGD-0000812
Ser N°	Configures the serial number of the instrument. Consult REGD-0000790

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Settings 2 Tab

This Tab is used to adjust the probe positioning in main pipetting areas (for more details consult REGD-0001221).

Buttons behavior:

Button	Behavior
Incubator	Start probe adjust procedure on incubator
Dilution cup	Start probe adjust procedure on dilution cup
Samples	Start probe adjust procedure on samples carousel
Reagents	Start probe adjust procedure on reagents carousel
Save Values	Saves X-Y settings in a file CalibValues.txt file (in Log folder). Overwrites existing CalibValues.txt file.

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Handler Tab

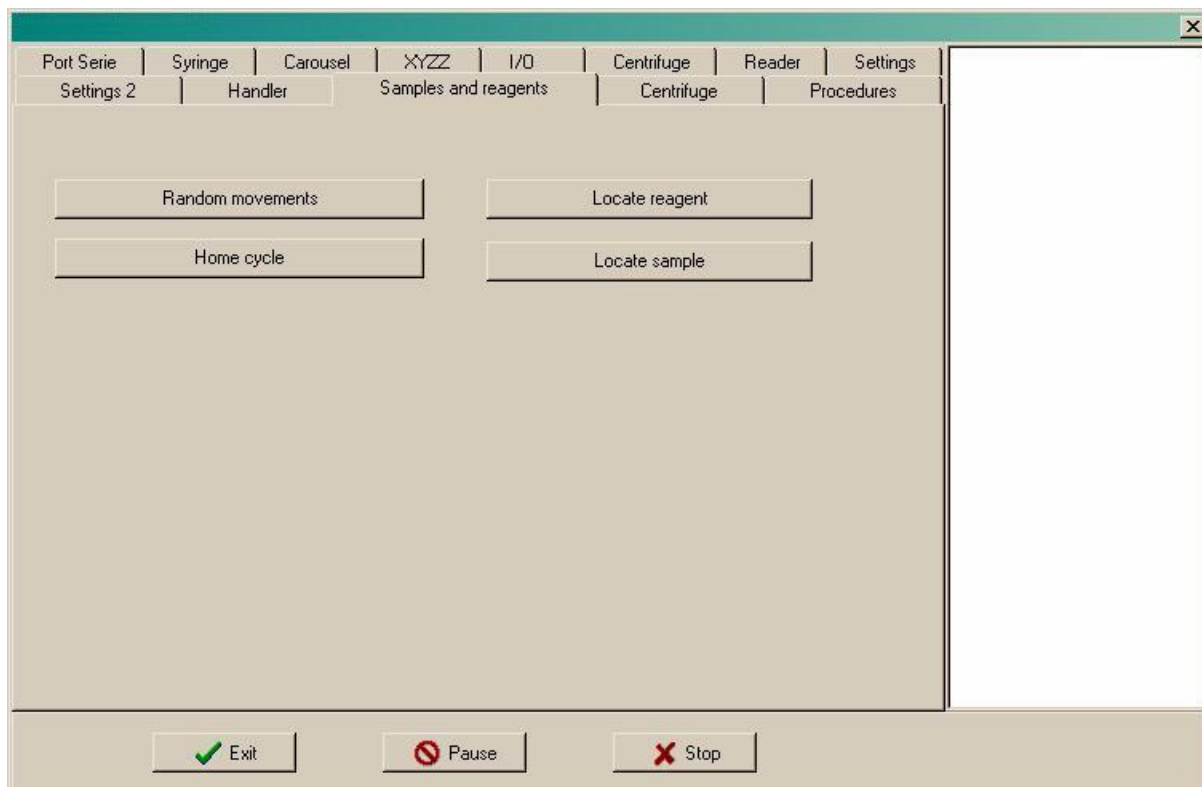
This Tab is used to check the handler.

Buttons behavior:

Button	Behavior
Random movements	Moves the handler motors randomly. Consult REGD-0000805
Level detection	Does a level detection procedure. Consult REGD-0000805
Sol. B integrity test	Perform a test to check the integrity of the Sol. B circuit. (This option is only available if the HW has been installed and the FluidicControls parameter of the Register System is enabled).
Impact detector	Perform impact of the probe over the incubator. The impact number is shown in the counter window and the counter can be reset.
Remove card from incubator	Removes a card from any of the incubator positions. Consult REGD-0000805
Place card into incubator	Places card into any of the incubator positions. Consult REGD-0000805
Remove card from centrifuge	Removes a card from any of the centrifuge positions.
Place card into centrifuge	Places card into any of the centrifuge positions.
Remove balancing card	Removes a card from any of the balancing positions.
Place balancing card	Places card into any of the balancing positions.

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Samples and reagents tab



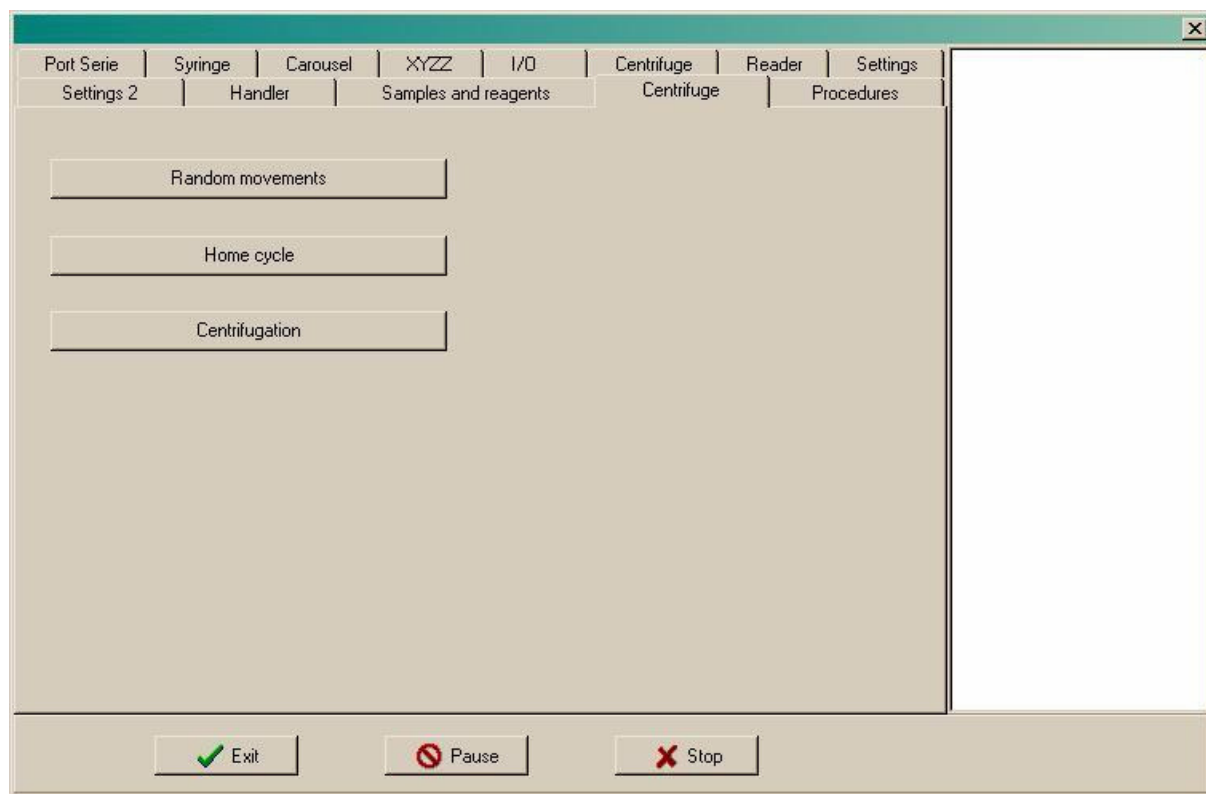
This Tab is used to check the samples and reagents carousels.

Buttons behavior:

Button	Behavior
Random movements	Moves the carousels randomly. Consult REGD-0000797
Home cycle	Does a cycle of homes. Consult REGD-0000797
Locate Reagent	Places a reagent in the pipetting position.
Locate sample	Places a sample in the pipetting position.

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Centrifuge Tab (High level)



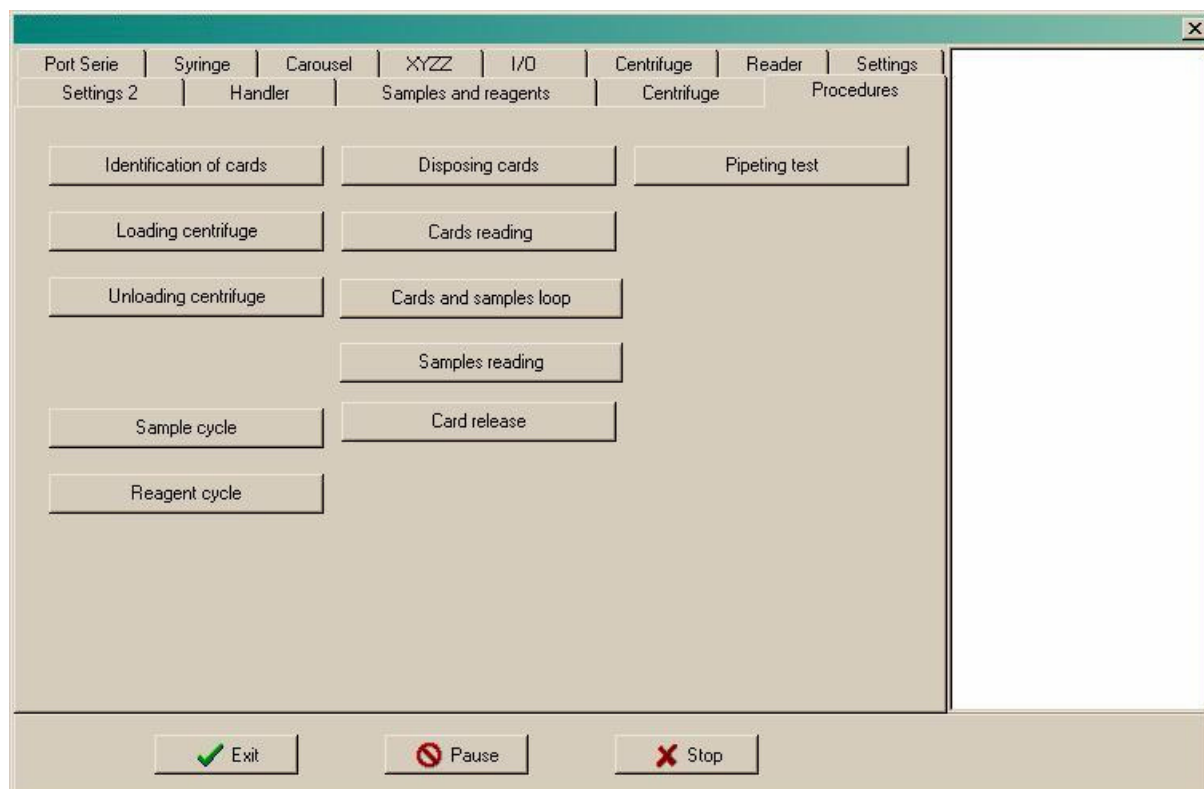
This Tab is used to check the centrifuge.

Buttons behavior:

Button	Behavior
Random movements	Moves the centrifuge randomly. Consult REGD-0000804
Home cycle	Does a cycle of home. Consult REGD-0000804
Centrifugation	Does a centrifugation cycle until "Stop" is pressed Consult REGD-0000804

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Procedures tab



This Tab is used to check the whole instrument.

Buttons behavior:

Button	Behavior
Identification of cards	Moves the cards to identification position.
Loading centrifuge	Starts a cycle loading the centrifuge, centrifuging and unloading it until "Stop" is pressed.
Unloading centrifuge	Unloads any cards from the centrifuge.
Samples cycle	Starts a cycle of detecting level on all samples. (Pumps should be switched on before starting the cycle).
Reagent cycle	Starts a cycle of detecting level on all reagents. (Pumps should be switched on before starting the cycle).
Disposing cards	Disposes all card to the card's trash.
Cards reading	Read all cards.
Cards and samples loop	Starts a cycle of read all cards and detect level on all samples.
Samples reading	Starts a cycle of reading all samples.
Card release	Starts a cycle to test card release.
Pipetting Test	Opens the pipetting test window. Consult REGD-0000783.

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
5.0	2005/07/12	<i>Document REGD-0000844_v5 replaces document 57074VOM002.</i>	Uploaded document in GxPharma.
6.0	2010/09/10	Updated technical software screenshots tabs. Included new settings 2 tab.	The document is updated due to the implementation of new functionalities.
7.0	2011/04/28	Changed Picture 3 in Procedure 1 – “NoInit” mode.	The Picture 3 was not of the “NoInit” mode.
7.0	2011/04/28	Added “Pumps should be switched on before starting the cycle” in Samples and Reagents Cycle button behavior.	The functionality of Sample and Reagent cycle requires the previous activation of the pressure and vacuum pumps.
7.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY-MM-DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
7.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
8.0	See document header.	Document REGD-0000844_v8 replaces document REGD-0000844_v7.	Document version changed.
8.0	See document header.	Document available for DG-57 and DG-517.	Error code list document reference deleted.
8.0	See document header.	Document title.	Commercial reference substituted by project reference.
8.0	See document header.	Document header.	Increase information to easy document search.